

MITHUUSAN KIRUPANANTHAN

Software Engineer | Full-Stack, Backend, and Cloud Applications
647-568-7527 | mithuuskank@gmail.com | LinkedIn | GitHub | Portfolio

Education

Toronto Metropolitan University

Sep 2021 – Apr 2025

Bachelor of Science (Honours), Computer Science

Toronto, ON

- **Coursework:** Algorithms, Data Structures, Databases, Operating Systems, Networks, Software Engineering

Work Experience

SGMC Canada

Jan 2026 – Present

Software Developer (Contract)

Toronto, ON

- Led development of Connect2Talents' initial web application using **Next.js**, **React**, **TypeScript**, and **Tailwind CSS**, delivering core onboarding, dashboard, and document review workflows.
- Built secure **Express.js** and **PostgreSQL** backend services with authentication APIs, role-based access control, protected routes, and **AWS S3** uploads, enabling secure onboarding and document handling.

Muia Consulting

Sep 2025 – Dec 2025

Software Engineer Intern

Toronto, ON

- Built a **FastAPI** backend using Google Gemini 2.0 Flash and Google Document AI to convert unstructured tax documents into structured SR&ED claim data, automating a key step in claim preparation.
- Designed an OCR + **Gemini** document pipeline with normalized **JSON/CSV** outputs, reducing SR&ED workflow time by **50%**.
- Developed **Python** services to process financial records, automate **ITC** eligibility checks, and enforce validation logic for accurate outputs.

Stephen Lewis Foundation

Oct 2024 – Dec 2025

Program Officer (Data & Analytics)

Toronto, ON

- Delivered a **Power BI** dashboard and automated intake workflow that cut manual reporting time by **50%** across **5 teams** in a **100+ employee** organization.
- Built **SQL**-based data pipelines with validation and transformation logic, replacing a days-long reporting process with instant access to decision-ready insights.

Equitable Bank

May 2022 – Dec 2022

Cloud & DevOps Engineer Intern

Toronto, ON

- Designed **CI/CD** pipelines to automate build, test, and deployment workflows, supporting **25+ production deployments** for a digital banking card project.
- Built an **API dependency map** to identify downstream service impacts faster and reduce downtime risk during weekly and monthly releases.

Relevant Projects

NovaPrep | React, Express.js, Amazon Bedrock, AWS Amplify, AWS App Runner

- Built and deployed **NovaPrep**, a live AI interview platform, using a React/Vite frontend and **Express.js** backend to support **50+ concurrent users**.
- Integrated **Amazon Nova Lite** and **Nova Sonic** through **Amazon Bedrock** for adaptive question generation, speech transcription, and answer evaluation.
- Deployed the application on **AWS Amplify** and **AWS App Runner** with automated **CI/CD**.

ForgeFit | React Native, TypeScript, Supabase, GPT-4o, USDA FoodData Central

- Developed a cross-platform **React Native** app for **iOS** and **Android** with workout tracking, nutrition logging, and health management features.
- Implemented AI-powered meal analysis using **GPT-4o** and **USDA FoodData Central**, providing personalized nutrition insights with **95%** macro accuracy.
- Integrated **Supabase** for authentication, session management, and cloud data storage, improving application reliability and enabling secure persistence of user workout and nutrition data.

Skills

Languages: Java, Python, TypeScript, SQL

Frameworks: React.js, Next.js, React Native, FastAPI, Express.js, Tailwind CSS

Databases/Cloud: PostgreSQL, Supabase, AWS, Docker, Jenkins, Git, Linux

Tools/AI: CI/CD, Power BI, OpenAI API, Google Document AI, Google Gemini, Amazon Bedrock

Concepts: REST APIs, Authentication, RBAC, Data Validation, Agile Development, SDLC, Debugging